

WAKE: a visitation-statistics atlas of the solar neighbourhood — arrival statistics, an exclusion map, and a flyby network on Gaia DR3 kinematics

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Abstract

Should we have been visited? We present the first calculation of visitation statistics for the solar system on the *observed* stellar kinematics of the solar neighbourhood: arrival statistics of close stellar passages on Gaia DR3 (56,286 stars with 6D phase-space data within 100 pc), combined with settlement dynamics in the probe-based, entry-driven convention of Carroll-Nellenback et al. Our contributions are three layers on one catalogue. (1) *Arrival statistics*: a completeness-corrected, exposure-normalized encounter rate with explicitly declared accounting conventions — reconstructing the accounting of the two published rate camps on our own data, we find their factor-of-two split is dominated by the population that the completeness correction targets, not by data quality; we also identify a third systematic mode (faint-end Gaia RVS radial-velocity artifacts) that evades both value- and error-based quarantines and can inflate encounter rates severalfold, and we introduce a tangential-velocity consistency test to flag it. Our clean rate is $\lambda(1\text{ pc}) = 4.5^{+4.0}_{-2.7}\text{ Myr}^{-1}$ for the FGK+early-M population. (2) *An exclusion map*: conditional statements of the form “if the settled fraction exceeds f^* , the solar system should have been visited within the past 10 Myr (silence rejected at 95%)”; at the standard probe range of 3.07 pc we obtain $f^* = 5.0 \times 10^{-3}$. A three-valued theorem layer imports the survival theory proved in a companion mathematics paper (DOI 10.5281/zenodo.21955413), separating where survival is theorem-guaranteed, where the theorem is silent, and recording that no extinction theorem exists for the full model anywhere. (3) *A flyby network*: on the time-varying graph of close passages, a propellant-minimal visiting route exists at all times in the window, with a single-transfer departure budget of $\Delta v \simeq 5.8\text{ km s}^{-1}$ — a statistical property of the network, not of any particular star. All statements are conditional probability statements over a declared population; undecidable regions are reported as undecidable. This is surveying, not proof.

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1 Introduction

Should we have been visited? If settled systems exist in the Galaxy and expand in the cheapest available channel — slow, small, and old probes launched at close stellar passages — then the question of whether the solar system has been reached is not a matter of speculation but of *measured kinematics*: which stars have passed within probe range, when, and at what rate, given the selection function of the survey that observed them. This paper computes those statistics on Gaia DR3 and overlays settlement dynamics on them.

This is the third branch of a visitation-hypothesis research programme (warp-class channels were treated in ASTROGATION, higher-dimensional channels in TESSERA); the present branch is the sub-light channel, for which the solar neighbourhood’s real kinematics are the decisive input. Our claim to novelty is deliberately narrow: *the overlay of settlement dynamics on real-catalogue encounter statistics*. The first layer alone — encounter statistics — is a well-developed field [2, 1, 9, 7]; we treat it as our verification bench, reproducing published anchors before adding anything (Section 3). The mathematical foundations of the settlement-dynamics layer (survival and extinction theorems for contact processes on ballistically moving points) were established in a companion mathematics paper [11]; here we import only its assumptions map and guarantees. Throughout, the discipline of the project constitution applies: every output statement is a conditional probability statement over a declared stellar population; regions where the data cannot decide are reported as *undecidable*, never silently filled; and the whole is surveying, not proof.

2 Data, selection, and quarantine

2.1 Catalogue and completeness

We use the Gaia DR3 six-dimensional sample within 100 pc: 56,286 stars with astrometry and radial velocities [8]. Selection is corrected by inverse-probability weighting (IPW) over the DR3 RVS selection function [4], with a floor $s_{\text{floor}} = 0.05$: stars with selection probability below the floor are not up-weighted but reported as an *undecidable population* (697 stars), so that the corrected population is explicitly FGK and early-to-mid M. Late-M and fainter populations are excluded from all rates and flagged as undecidable (a bounded-rate extension is reserved but not used here).

2.2 Five quarantine systems and a third failure mode

Individual-event claims are protected by quarantine flags: astrometric quality, proper-motion signal-to-noise, extreme radial velocities ($|RV| > 550 \text{ km s}^{-1}$, with a whitelist path for confirmed hypervelocity stars), and a radial-velocity error cut ($\sigma_{RV} > 20 \text{ km s}^{-1}$ excluded from individual event judgements while still contributing to rates).

During the accounting cross-checks of Section 4 we identified a *third* systematic mode that evades both the value-based and the error-based quarantines: **faint-end RVS radial-velocity artifacts** — stars at $G \gtrsim 14$ with $|RV| = 150\text{--}900 \text{ km s}^{-1}$ but disk-like tangential motion. Genuine halo stars show halo kinematics in *both* components; a disk-like tangential velocity with an extreme radial velocity is the artifact signature (cf. the DR3 RV quality guidance of [10]). We therefore add quarantine bit 5 ($G > 13.5 \wedge |RV| > 150 \text{ km s}^{-1}$; 1,323 stars; threshold scanned over $120/150/200 \text{ km s}^{-1}$ with a stable clean component), operated like the hypervelocity bit: excluded from individual event judgements, rates reported *both ways* (clean and suspect-inclusive), manual review by the tangential-velocity consistency test, and a rescue whitelist (empty after review: every top suspect showed the artifact signature, e.g. $RV = -790 \text{ km s}^{-1}$ with $v_{\text{tan}} = 18 \text{ km s}^{-1}$). Left unquarantined, this population would inflate the sub-parsec encounter rate by a factor of ~ 7 (Section 4); we flag this as a general methodological caution for DR3-based encounter studies.

3 Propagation and verification

Orbits are propagated in an axisymmetric Galactic potential (MWPotential2014-class) by direct numerical integration; an independent epicyclic analytic path serves as the formulation-independence counter-path (gate G2). Verification anchors (gate G3) reproduce published encounters before any new statistics are claimed: the Scholz-star flyby [12, 6], GJ 710 (our value 1.294 Myr , 0.0503 pc , in agreement with [5]; the published-value split among DR3 analyses is documented neutrally in Appendix A), and the encounter-frequency anchor [2]. A dual-potential audit (McMillan17 [13] via a spline tabulation whose interpolation error is itself audited, max 3.7×10^{-5} relative) bounds potential systematics; where the two-potential difference exceeds a star’s measurement uncertainty at far epochs, the star’s *effective horizon* is shortened (“ t_{pot} ”, a third horizon component), so that model uncertainty is treated exactly like measurement uncertainty: as undecidability, not as noise to be averaged.

Each star carries an *individual horizon* t_h : the epoch beyond which its position uncertainty (dominated by the radial-velocity error, $1 \text{ km s}^{-1} \approx 1 \text{ pc Myr}^{-1}$) exceeds half the probe range. Statistics are computed in the *control region* $|t_{\text{ph}}| \leq t_h$; events beyond horizons form the undecidable shadow visible in Figure 1 — the measured “evidence horizon” of the catalogue. Two

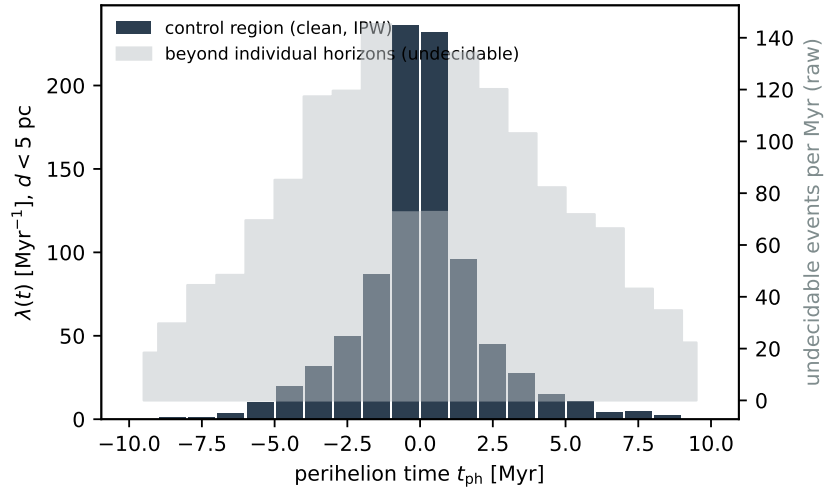


Figure 1: Exposure-normalized arrival rate $\lambda(t)$ within 5 pc for the clean population (bars), with the undecidable shadow of events beyond individual horizons (gray). The envelope is the measured evidence horizon of Gaia DR3: rates are only claimable where the catalogue can support them. Reproduce: `python3 src/wake_p5/paper_figs_main.py`.

implementation defects found and fixed by the G2 gate during this work (a parabolic-refinement coefficient and a sharp-minimum breakdown, repaired by exact d^2 -interpolation; all G3 anchors re-passed) are disclosed in Appendix B.

4 Arrival statistics and the accounting of encounter rates

4.1 Exposure accounting

A rate is a count divided by an exposure, and the choice of exposure is a convention that must be declared. We normalize each star’s contribution by its own effective exposure $2 \min(T, t_h)$ within the control region. A naive uniform window ($1/2T$) dilutes the rate as T grows past typical horizons; our own first-look value of 11.7 Myr^{-1} at 1 pc arose from exactly that convention (horizon-ignored, uniform-window) and is reproduced exactly under it — a downward-biased accounting, not a different sky. The exposure-normalized clean rates are

$$\lambda(1 \text{ pc}) = 4.49 [1.78, 8.50] \text{ Myr}^{-1}, \quad \lambda(2 \text{ pc}) = 27.5 \text{ Myr}^{-1}, \quad \lambda(5 \text{ pc}) = 137.3 \text{ Myr}^{-1}$$

(95% star-bootstrap interval at 1 pc; population: FGK+early-M, IPW-corrected; suspect-inclusive values are co-reported in the catalogue). The rate scaling departs from the geometric d^2 law ($6.1 \times$ from 1 to 2 pc, $5.0 \times$ from 2 to 5 pc), reflecting heavy-tailed, few-star-dominated contributions and the distance structure of the IPW weights; the d^2 -law tests of Section 3 live in the BJ-convention reconstruction, a different accounting.

4.2 Three conventions on one data set

The two published 1-pc rate camps — $19.7 \pm 2.2 \text{ Myr}^{-1}$ [2] and $10.6 \pm 4.5 \text{ Myr}^{-1}$ [7] — use different accounting conventions (window, exposure, completeness target, and population). We

reconstructed both conventions from the published texts (the reconstruction limits — five points for [2], seven for [7] that could not be determined from the texts — are listed neutrally in Appendix A) and applied all three to our single data set:

Convention (on our data)	$\lambda(1 \text{ pc}) [\text{Myr}^{-1}]$	Published
Ours (control region, IPW, per-star exposure)	$4.5^{+4.0}_{-2.7}$	—
BJ+18 reconstruction ($G \leq 12.5$ subset)	23.7 ± 2.6	19.7 ± 2.2
FP26 reconstruction (discrete, reliability cut)	21.3	10.6 ± 4.5

The BJ-convention value is reproduced on clean DR3 data — so residual spurious astrometry is unlikely to be the main driver of the published split. The remaining FP26 gap closes when the faint-end RV artifacts of Section 2.2 are excluded (bright-only discrete counting gives $\simeq 7.6$, matching their solar value of 9.1). Based on this reconstruction, the split is dominated by the *population that the completeness correction targets*: [2] corrects to the full mock stellar population, [7] does not correct beyond a window factor. Consistently, a two-ledger check on our own data — correcting by per-star exposure and IPW (denominator ledger) versus by a mock-based $C(t, d)$ completeness map (numerator ledger, Appendix C) — yields a measured *population bridge* of $5.3\times$ between the FGK+early-M and full-population rates, compared with the $\sim 9\times$ implied by the mock extrapolation of [2]; the difference is a quantitative basis for critiquing mock-based completeness corrections.

4.3 Arrival catalogue and structure

The arrival-event catalogue (2,197 stars with median perihelion within 5 pc inside $|t| \leq 13 \text{ Myr}$) reproduces the well-known close passages — HD 7977 (0.037 pc, -2.76 Myr), GJ 710 (0.052 pc, $+1.29 \text{ Myr}$) — and carries all quarantine, horizon, and both-ways rate metadata. The time-resolved rate $\lambda(t)$ shows no kinematically coherent clumps (all apparent clumps are single-star dominated; pairwise $\Delta UVW \geq 45 \text{ km s}^{-1}$), and the small approaching-vs-receding excess ($+2.1\%$ relative) is the textbook solar-apex dipole [15] (approaching fraction 75.4% toward the apex hemisphere vs. 24.1% away); after subtracting the sample mean-field reflex it reduces to $+0.78\%$ (1.9σ) — consistent with no spatial structure.

5 The exclusion map

5.1 Definition

The map’s conditional statement is defined by a Poisson visit count:

$$N_{\text{vis}} = f \cdot \frac{\langle \hat{p} \rangle}{f} \cdot \lambda(R) \cdot T_{\text{eff}}, \quad N_{\text{vis}} \geq 3 \iff P(\geq 1 \text{ visit}) \geq 95\%, \quad (1)$$

with $T_{\text{eff}} = \max(0, 10 \text{ Myr} - R/v_p)$ the support-limited retrospective window (probe travel time deducted), f the settled fraction, $\langle \hat{p} \rangle / f$ the mark-field factor (unity for the constant field; evaluated on the real arrival kinematics for the front family), and $\lambda(R)$ the clean-population rate interpolated between measured anchors (below 1 pc: d^2 extrapolation, flagged; above 5 pc: undecidable). The clean rate is a *lower bound* on the physical encounter rate (probes do not obey Gaia’s selection function), so the “should-have-been-visited” region is conservatively small; a

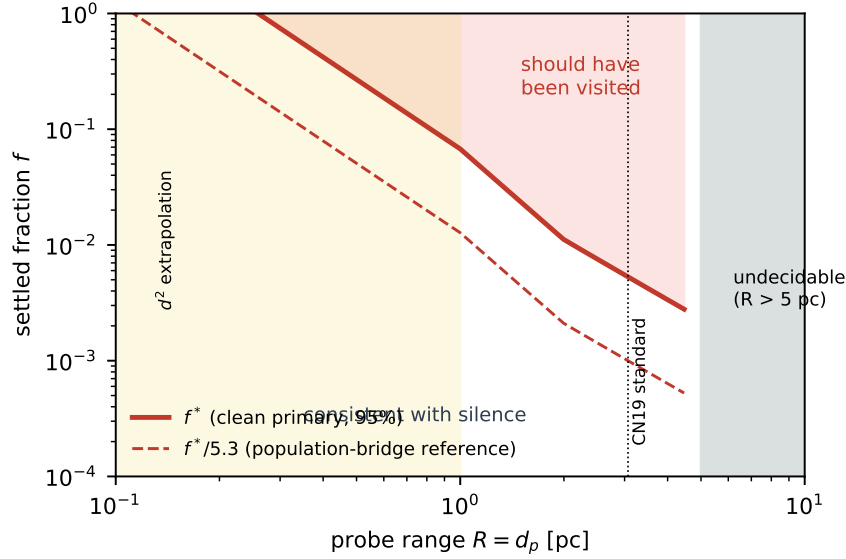


Figure 2: The exclusion map in the (R, f) plane at $v_p = 10 \text{ km s}^{-1}$ (constant mark field). Above f^* (solid), silence is rejected at 95% for the declared population; the dashed line is the population-bridge reference. Gray: undecidable ($R > 5 \text{ pc}$); yellow band: d^2 -extrapolated. At the standard probe range $R = 3.07 \text{ pc}$: $f^* = 5.0 \times 10^{-3}$.

population-bridge reference layer ($\times 5.3$, model-explicit) and the suspect-inclusive layer are co-reported.

At the standard probe range of [3] ($R = 3.07 \text{ pc}$),

$$f^* = 5.0 \times 10^{-3}$$

— if more than half a percent of systems in the declared population are settled (and launch on entry), the solar system should have been visited within the past 10 Myr. At $R = 1 \text{ pc}$, $f^* = 6.8 \times 10^{-2}$ (Figure 2); boundary uncertainty follows the rate interval ($\times 0.53$ – $\times 2.52$).

5.2 The three-valued theorem layer

The settlement-dynamics layer imports the survival theory of [11] with its assumptions, not beyond them. The assumptions map (mixing number $K_{\text{mix}} = w_1 T_s / R \geq 20$; bounded relative density $\bar{\Lambda} \leq 4$ on a speed shell, which the isotropized DR3 distribution satisfies non-trivially at $\bar{\Lambda} = 3.3$ – 3.6) yields a **three-valued legend**: (i) *C2-guaranteed* (assumptions hold and the numerical threshold is exceeded), (ii) *theorem-silent* (assumptions fail — notably the *entire* standard scan domain of [3], $R = 3.07 \text{ pc}$ with $T_s = 0.1$ – 1 Myr , is theorem-silent; guarantees open only at $T_s \gtrsim 3 \text{ Myr}$), and (iii) the reminder that *no extinction theorem exists for the full model anywhere* (the companion paper proves the SIR-type bound does not extend), so “settlement impossible” is nowhere theorem-backed. The asymmetry is physical: for short-lived settlements the mathematics is silent; the longer settlements live, the more strongly silence constrains them. Where the theorem is silent the only judgement material is the numerical threshold, which we *re-calibrated on the measured kinematics*: the survival threshold shifts from the isotropic calibration $m \approx 1.0$ – 1.3 to $m \approx 1.5$ – 2.0 (Figure 3) — real anisotropic kinematics are *harsher* to settlement

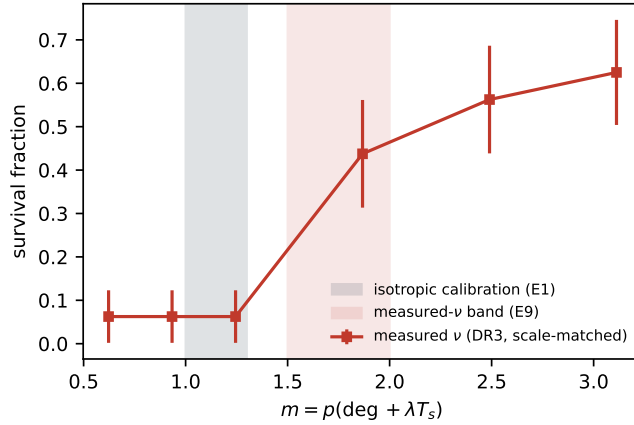


Figure 3: Survival threshold on measured DR3 kinematics (scale-matched to the isotropic calibration): the threshold band shifts upward by $\sim 1.5\times$. The exclusion map’s guarantee layer uses the conservative edge ($m > 2.0$).

fronts than the isotropic approximation, which gives the pure-isotropy open problem of [11] an observational demand.

5.3 Front-direction sensitivity

For the front-family mark field, the direction average over two priors (uniform sphere vs. galactocentric bias) shifts the map by only 2.5% — prior-robust. The *per-direction* spread is large (52%): arrivals are apex-concentrated, so a settlement front’s effectiveness depends on its direction relative to the solar apex — a direction dependence we report as a finding connecting to the settlement-direction discussion of [17].

6 The flyby network

On the time-varying graph of clean solar passages (646 nodes within horizons, perihelion states extracted by the numerical propagator), we ask the constitution’s third-layer question: what is the propellant-minimal visiting route? A visitor riding any passing star may transfer, by one ballistic cruise leg, to a star whose perihelion is arbitrarily deep in the solar system. The minimal single-transfer departure budget is

$$\Delta v \simeq 5.8 \text{ km s}^{-1}$$

(5.93 km s^{-1} for the deepest target $d \leq 0.1 \text{ pc}$; 5.85 km s^{-1} for $d \leq 1 \text{ pc}$) — and this is a *statistical property of the network*, not of a particular star: over the error-MC ensemble the route exists in 100% of draws with median $5.7\text{--}5.9 \text{ km s}^{-1}$ (departure charged entirely to propellant — co-moving riders cannot use flybys to accelerate — so the budget is an upper bound). Direct passages within 0.1 pc in the window are HD 7977 and GJ 710, matching the known record. The graph estimate is validated by a formulation-independent counter-path: Jacobian shooting in the full potential converges to sub- 0.1-mpc misses and shifts the transfer budget by at most 0.19 km s^{-1} (tidal terms are subdominant). Multi-leg optimization, full ensemble re-extraction, and real stellar masses are reserved for v1.1 of the network file.

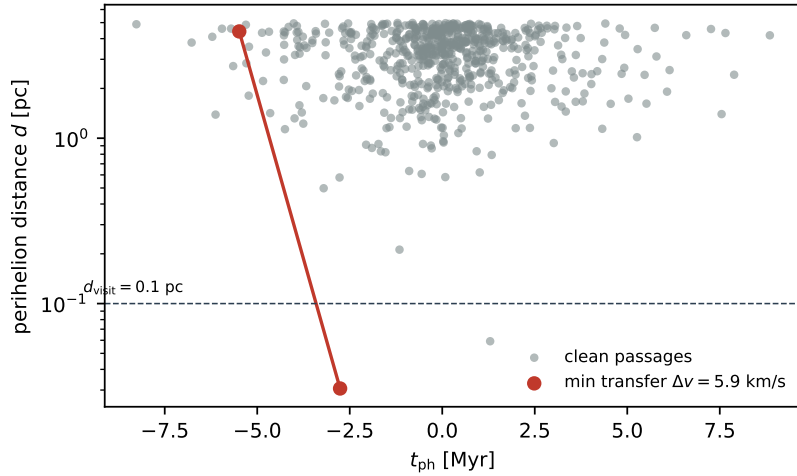


Figure 4: The flyby network’s node set (clean passages; perihelion distance vs. time) and the minimal single-transfer route into $d \leq 0.1$ pc.

7 Discussion

The philosophy of undecidability. The atlas’s reliability rests on an asymmetry principle — prefer undecidable over falsely decidable — implemented in three mechanisms: individual horizons with three components (measurement growth, potential-model divergence, budget-type truncation), five quarantine systems with both-ways reporting, and the three-valued theorem layer. Nothing in the map is painted by silent extrapolation.

Limitations. The theorem layer judges under an isotropized approximation of the measured velocity distribution (the real distribution is anisotropic; E9 quantifies the direction of the error). The flyby network assumes solar-mass deflectors and single transfers. The d^2 extrapolation below 1 pc and everything above 5 pc are flagged. The population is FGK+early-M; late-M visitation channels are undecidable here (a bounded-rate extension is reserved).

Comparisons with published values rest on our reconstruction of published accounting conventions; the points that could not be determined from the texts are listed openly in Appendix A. DR4 replacement is designed in from the start (the catalogue is a swappable data layer).

8 Conclusions

On the observed kinematics of the solar neighbourhood, with a declared population and declared accounting: (1) the clean arrival rate at 1 pc is $4.5^{+4.0}_{-2.7} \text{ Myr}^{-1}$, and the published factor-of-two split is dominated by completeness-target population choices, not data quality; (2) if more than 0.5% of systems (standard probe range) are settled, the solar system should have been visited within the past 10 Myr — silence is informative at 95% above that line, and the mathematics guarantees survival of settlement processes only for long-lived settlements, so silence constrains long-lived settlement most strongly; (3) a propellant-minimal visiting route through the real stellar traffic exists at $\Delta v \simeq 5.8 \text{ km s}^{-1}$ at all times. Methodologically we contribute the exposure-accounting analysis, the faint-end RV artifact quarantine with its tangential-velocity test, and the two-ledger

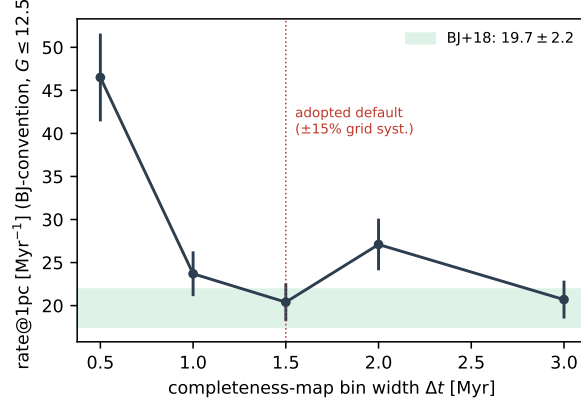


Figure 5: Bin-width scan of the mock-completeness rate reconstruction: a hidden regularization parameter, disclosed.

completeness check with its measured population bridge. The atlas is surveying, not proof — but it turns the visitation question into numbers that Gaia DR4 can sharpen.

A GJ 710, published-value splits, and reconstruction limits

[GJ 710 perihelion values across DR2/DR3 analyses; our reproduction (1.294 Myr, 0.0503 pc) agrees with [5]; a unit-conversion hypothesis for one published time value is stated neutrally (our 1.294 Myr expressed in pc/(km/s) units equals 1.3234, coinciding with the published 1.324 ± 0.026 to 0.1%). The full accounting comparison table for [2] and [7], with the twelve points that could not be determined from the published texts, is reproduced here from the repository records; these lists disclose the limits of our reconstruction.]

B Method record and AI disclosure

[Complete verification protocol: claim-driven falsification against an independent simulation device; machine certification of geometric constants; error ledger including the two propagation-refinement defects found by gate G2 and the “agreeing numbers are the ones to distrust” episode (two mutually confirming rate values, 11.7 and 20.3, were both wrong for different reasons and were caught by convention identification and engine re-verification). Roles of AI systems (Claude Code, model Claude Fable 5) as executor under human arbitration; steering-intervention log empty; this internal verification is not a substitute for human peer review.]

C Completeness-map bin width as a hidden regularizer

[The mock-based $C(t, d)$ completeness map [14, 16]: its bin width acts as a hidden regularization parameter: the reconstructed BJ-convention rate scans non-monotonically ($46.5 \rightarrow 23.7 \rightarrow 20.4 \rightarrow 27.1 \rightarrow 20.7 \text{ Myr}^{-1}$ for $\Delta t = 0.5 \rightarrow 3.0 \text{ Myr}$; Figure 5) — fine bins diverge through empty-cell $1/C$ weights, coarse bins alias. We adopt $(\Delta t, \Delta d) = (1.5 \text{ Myr}, 0.75 \text{ pc})$ with a $\pm 15\%$ grid-systematic band, and co-report the canonical (1.0, 0.5) value.]

D Reproducibility and data release

[The public data set: `arrival_catalog_v1.json`, `exclusion_map_v1.json` (v1.0.1), `flyby_network_v1.json`, with a SHA-256 manifest; regeneration commands for every figure and table; the browser simulator reads exactly these files (additive-compatibility schema principle). This appendix is the paper-side anchor of the three-artifact consistency audit (paper, data/maps, simulator).]

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